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Electoral reform and party-system fragmentation: diminishing marginal effects and vote–seat conversion limits in Chile’s concurrent local elections

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ABSTRACT

Evidence on the impact of electoral reform on party systems in new democracies is limited. This study analyses 1,725 Chilean municipal contests (2004–2021), leveraging automatic, population-triggered increases in council district magnitude and the concurrent timing of first-past-the-post mayoral and proportional representation council elections. Using a difference-in-differences design and mixed-effects models, larger district magnitudes are found to increase the effective number of vote-winning parties, though the marginal effect diminishes at higher magnitudes. Concurrent FPTP elections heighten vote-level fragmentation but not seat-level fragmentation, reflecting conversion limits under proportional representation. Institutional effects materialise gradually over several electoral cycles. The findings show that institutional design exerts an independent effect on party-system fragmentation beyond social structures, extending the institutions-first logic to a subnational PR setting in a new democracy and showing how automatic seat adjustments provide causal leverage for identifying institutional effects.

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KEYWORDS Electoral reform; district magnitude; electoral systems; local elections; party system fragmentation; new democracies

1. Introduction

Understanding the causal impact of electoral reforms is particularly crucial in new democracies, where institutional changes can produce lasting effects on the political landscape. In these settings, where democratic institutions continue to evolve, modifications to electoral rules may alter party systems, voter behaviour, and overall democratic stability (Brinks, Levitsky, and Murillo 2020). Unlike consolidated democracies, new democracies – systems where

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democratic institutions are recently established and still consolidating – are characterised by more frequent institutional reform and greater unevenness in how electoral rules are implemented across regions, which can produce divergent outcomes in party system development (Calvo and Negretto 2020). The central question this raises is whether institutional rules exert an independent and identifiable effect on party competition in such contexts – and whether that effect follows the same patterns that theory predicts in more stable settings.

Prior work suggests there is reason to expect institutional logic to operate in qualified form rather than as classical theory predicts. Interparty competition stabilises only after long runs of competitive elections (Mainwaring and Zoco 2007), the same electoral rule generates different pressures depending on political context (Moser and Scheiner 2012), and rule instability itself dampens institutional discipline on the party system (Bunker and Negretto 2024). Yet systematic evidence on whether and how these qualifications hold in new democracies remains scarce, and this study addresses that gap directly.

Two families of explanation compete in this literature. Institutional theories argue that electoral rules are the primary drivers of party system fragmentation (Duverger 1954; Lijphart 1990; Taagepera 2007). Social theories, by contrast, attribute fragmentation to underlying cleavage structures and ethnic heterogeneity (Lipset and Rokkan 1967; Ordeshook and Shvetsova 1994). The conventional resolution treats the two as complementary, with social structure providing the raw material and institutions shaping how it is translated into party competition (Clark and Golder 2006; Neto and Cox 1997). This study contributes to that debate by testing the institutional account at the district level in a new democracy, examining how district magnitude shapes party-system fragmentation across permissive rules and how vote- and seat-level fragmentation relate under concurrent electoral arrangements.

To understand these dynamics, the study turns to district-level data, where institutional variation and electoral behaviour can be directly observed. While national analyses offer valuable insights, district-level studies provide a more precise view of how electoral reforms shape political competition in specific contexts. Districts vary in population size, demographic composition, and political environment, which makes them suitable units for assessing institutional effects (Singer 2013; Singer and Gershman 2018). Focusing on the district level helps capture the localised impact of reforms, such as changes in district magnitude, and how these reforms influence the effective number of parties (ENP).

Within the subnational framework, this study examines the interaction between institutions and party systems in Chile, a new democracy that offers valuable variation in local electoral design. It focuses on municipal elections, which are conducted through concurrent local contests: voters cast ballots in

separate first-past-the-post (FPTP) elections for mayors and proportional representation (PR) elections for local councillors held on the same day. Local elections are not merely a convenient methodological setting; they constitute a substantively distinct arena of political competition, with lower electoral salience, stronger incumbency effects, and a more direct link between population size and institutional design than national contests (Gómez Díaz 2021; Tiemann 2015). Because the size of council districts adjusts automatically as populations increase or decline, this setting provides an ideal opportunity to analyse how institutional adjustments influence party competition at the local level. The concurrent structure of these elections also allows the joint examination of centripetal and centrifugal incentives within the party system (Calvo and Hellwig 2011).

While the subnational approach allows us to leverage the causal mechanisms linking institutions and party system fragmentation, examining concurrent local elections helps isolate potential coattail effects and other confounding factors that might distort the underlying patterns of party competition.

The remainder of this article is structured as follows. [Section 2](#) reviews the theoretical framework. [Section 3](#) presents the Chilean case as an ideal setting to assess the shortcomings in the literature and details the local electoral system and its institutional effects. [Section 4](#) describes the research objectives and outlines the hypotheses. [Section 5](#) presents the data and methods. [Section 6](#) reports the findings, emphasising how changes in district magnitude influence the effective number of political parties. [Section 7](#) discusses the broader implications and [Section 8](#) offers concluding remarks.

2. Theory

There are several key theories that seek to explain how party systems develop and change over time, and their relative importance remains contested. One of the most influential of these focuses on the role of institutional features, such as district magnitude (Duverger 1954), the type of electoral formula used (Lijphart 1990), and the thresholds for representation (Gallagher 1991; Rae 1967).

The core idea is that electoral systems with more permissive rules – meaning those with higher district magnitudes and proportional-representation formulas – are more likely to produce fragmented party systems. These systems lower the barriers for smaller parties to gain seats, allowing more parties to enter and persist in the political arena. In contrast, systems with lower district magnitudes or majoritarian principles, such as first-past-the-post, tend to yield less fragmented party systems because they give a disproportionate advantage to larger, more established parties and make

it harder for smaller ones to win representation (Bochsler 2022; Carey and Hix 2011; Hicken and Stoll 2017).

Recent studies have refined this view, arguing that the number of available seats alone can provide a reliable prediction of how many parties will compete in an election (Taagepera 2007). In this minimalist, institutions-only approach, district magnitude and assembly size can be used separately or together, depending on the level of analysis, to forecast the number of parties likely to win votes and seats (Li and Shugart 2016; Shugart and Taagepera 2017). These updates demonstrate that institutional effects remain robust across a wide range of contexts, including both established and newer democracies (Lago and Martínez i Coma 2024; Lucardi 2019).

A rival theory suggests that fragmentation is driven less by institutional design and more by social structures. Early research emphasised societal cleavages as the main dividing forces (Lipset and Rokkan 1967). Later work highlights that social heterogeneity and ethnic diversity also contribute to fragmentation (Moser, Scheiner, and Stoll 2018; Ordeshook and Shvetsova 1994; Powell 1982). The general understanding is that in societies with higher levels of ethnic or social division, parties are more likely to form around specific interests, producing larger and more numerous party systems (Birnie 2006; Madrid 2005; Stoll 2008). Importantly, however, this effect is conditional. Bochsler (2011) shows that the positive impact of ethnic diversity on fragmentation holds only when a group is large enough and territorially concentrated, rather than dispersed across a country.

The conventional resolution to this debate is that the two explanations are not competing but complementary (Cox 1997; Neto and Cox 1997). In this view, social structure provides the underlying source of political division, while electoral institutions mediate how those divisions are translated into party competition (Clark and Golder 2006; Teorell and Lindstedt 2010). More recent district-level studies explore how institutional permissiveness interacts with social heterogeneity, showing that constituency diversity can amplify fragmentation under permissive rules (Bunker 2025a; Potter 2014). Generally, institutional analyses confirm that the influence of diversity on fragmentation tends to increase as the number of available seats grows, while restrictive systems dampen it (Harfst et al. 2018; Singer and Gershman 2018).

Although extensive evidence suggests that each of these theoretical accounts is at least partially correct, most supporting research comes from consolidated democracies, and findings from new democracies remain scarce and inconsistent. This imbalance underscores the need to reassess institutional and integrative explanations in contexts where democratic institutions are still evolving and electoral reforms occur more frequently.

Building on these observations, scholars have begun to address such limitations by analysing elections at more granular levels (Harfst et al. 2018; Potter 2014; Raymond 2015; Singer and Gershman 2018; Tiemann 2015),

seeking to link institutional reforms to observable electoral outcomes (Banerjee and Rich 2016; Lago and Martínez i Coma 2024). However, comparable research in new democracies has progressed more slowly (see Bunker 2025a; Gómez Díaz 2022; Lucardi 2016, 2019; Maškarinec 2026; Michalak 2023).

A further open question concerns the temporal dimension of institutional effects. Taagepera (2002) argues that electoral outcomes align with institutional expectations only gradually, after voters and parties have had several cycles to learn and coordinate under new rules. Research in new democracies and at subnational levels confirms that this adjustment process can be slow and uneven (Bunker and Negretto 2024; Roussias 2022; Tavits 2005; Tavits and Annus 2006).

3. The Chilean case

Chile provides particularly useful leverage for analysing institutional effects on party-system fragmentation in a new democracy setting. Automatic adjustments in district magnitude as populations change generate exogenous institutional variation that is independent of political dynamics, allowing electoral reform to be treated as an exogenous treatment in a natural experiment. The concurrent structure of local elections, where first-past-the-post contests for mayors coincide with proportional-representation contests for councillors, further allows simultaneous examination of centripetal and centrifugal incentives, and helps isolate confounders that could otherwise distort the association between institutional design and party competition.

Chile's series of consecutive local elections under the same basic framework, combined with periodic automatic adjustments to district magnitude, also provides an opportunity to examine whether and how quickly institutional effects materialise over time. While prior work (such as Bunker 2025a) demonstrates that institutional and social factors can interact under these conditions, the independent effect of district magnitude, the limits of vote-to-seat conversion, and the temporal pace at which institutional effects materialise have only been partially explored.

Chilean local elections, held since 1992, are a key component of the country's political system. They are conducted every four years and involve the simultaneous election of mayors (*alcaldes*) and local councillors (*concejales*). The mayoral elections are decided using a first-past-the-post (FPTP) system, where the candidate with the most votes wins. In contrast, the local councillor elections use a proportional representation (PR) system. This makes the system a concurrent dual-ballot arrangement, with voters casting two separate ballots while parties face both centripetal and centrifugal incentives at the same time (Bunker 2008).

Both mayors and local councillors are elected to serve in municipalities at the communal level. There are 345 communes, which correspond to the districts used in this analysis. The number of councillor seats in each district depends on the size of its population: smaller municipalities elect six councillors, medium-sized ones elect eight, and the largest elect ten. This structure ensures that council size is proportional to population, maintaining representational balance across districts (Montecinos 2007).

The election of local councillors is carried out using the D'Hondt method of proportional representation. In this system, votes are divided by a series of divisors, and seats are allocated to the candidates from parties or coalitions with the highest quotients. This procedure distributes seats among parties in proportion to their vote shares, ensuring that representation reflects electoral support as closely as possible. No formal legal threshold applies in Chilean councillor elections; however, the effective threshold under D'Hondt approximates $1/(M + 1)$, implying barriers of roughly 14%, 11%, and 9% for districts electing six, eight, and ten councillors respectively (see Lijphart 1990).

During the period from 2004 to 2021, several automatic adjustments were implemented to update the number of councillor seats in municipalities whose populations had grown. These revisions increased the number of seats in most districts, with only one case of reduction. Despite these changes, the electoral framework remained stable: proportional representation under the D'Hondt method continued to be used, mayoral elections remained concurrent, and the total number of districts held constant at 345.

The stability of Chile's local electoral system across this period, combined with periodic automatic adjustments to district magnitude, holds the institutional framework constant within an otherwise evolving political environment.

4. Research objectives and hypotheses

The objective of this study is to extend insights from research on consolidated democracies to new democracies, building on the leverage that the Chilean case provides. Considering that social heterogeneity is known to influence fragmentation under certain conditions, it becomes especially relevant to explore the effects of institutional design in isolation. Specifically, the study evaluates the validity of institutional explanations at the district level by testing hypotheses derived from institutional theory in the context of local elections, where institutional variation is exogenous and directly observable.

Turning to the theoretical expectations, the first hypothesis posits that the effective number of parties winning votes (ENPv) will increase at a greater rate in districts that experience automatic seat increases compared with those that do not. Since the majority of districts remain unchanged, those subjected to reform can be treated as the treatment group in a quasi-experimental framework (Fiva and Hix 2021).

H1: Increases in district magnitude are expected to lead to a higher effective number of vote-winning parties (ENPv) in Chilean local councillor elections. This relationship should be particularly strong in districts that undergo automatic seat increases, such as changes from six to eight or from eight to ten seats, and is expected to become more evident over time as the effects of these reforms accumulate.

The idea is that as district magnitude rises, the effective threshold for representation falls, so more parties can plausibly secure a seat and voters waste fewer ballots on smaller options. Each successive seat added, however, represents a smaller proportional increase in permissiveness, so the marginal effect on fragmentation should taper as council size grows.

The second hypothesis suggests that the effective number of vote-winning parties (ENPv) is shaped by concurrent first-past-the-post (FPTP) elections for mayors. When elections with centripetal incentives, such as mayoral contests, occur simultaneously with elections featuring centrifugal incentives, such as local councillor races, the permissiveness of the proportional-representation (PR) election is conditioned (Calvo and Hellwig 2011; Riera 2013). In other words, concurrent FPTP elections are expected to increase vote-level fragmentation but limit its translation into seats due to conversion mechanics.

H2: Concurrency between first-past-the-post (FPTP) mayoral and proportional-representation (PR) council elections is expected to increase the effective number of vote-winning parties (ENPv) but not the effective number of seat-winning parties (ENPs). The simultaneous timing of elections introduces centripetal incentives that constrain the translation of vote-level fragmentation into seat-level fragmentation.

The logic plays out through the dual-ballot architecture, where voters can split tickets across the FPTP mayoral race and the PR council race held the same day. The centripetal pull of the mayoral contest does not foreclose the centrifugal incentives of the council contest, so vote-level fragmentation in council races is expected to rise under concurrency. D'Hondt allocation and the effective PR threshold, however, should absorb part of that fragmentation at the seat-allocation stage, so seat-level fragmentation should not rise commensurately.

5. Data and methods

5.1. Data

The analysis draws on the Chilean Political Landscape Dataset (CPLD), a standardised longitudinal dataset of district-level electoral results and

indicators covering all elections in Chile between 1989 and 2025 (Bunker 2025b, 2025c). For this study, we use the subset corresponding to 3,450 district-level contests in local elections. Of these, 1,725 correspond to local councillor elections and 1,725 to mayoral elections. Because these contests are held concurrently, variables are computed independently for each race to preserve the distinct dynamics of first-past-the-post (FPTP) and proportional representation (PR) systems.

The dependent variable is the effective number of parties (ENP), a standard measure introduced by Laakso and Taagepera (1979) to capture the number of relevant political parties in a system by weighting each party's vote or seat share. It can be calculated using either votes (ENPv) or seats (ENPs). The following formula shows the specification,

$$enp = \frac{1}{\sum_{i=1}^n p_i^2}$$

where p is the proportion (or percentage) of votes or seats that party i has, and n is the total number of parties. Because there is a mechanical effect between the number of parties and how votes are distributed (Benoit 2002; Clark and Golder 2006; Fiva and Folke 2016), it is understood that ENPs is constrained by ENPv. Thus, we begin the analysis focusing on the latter, ENPv, which can be conceived as the primary determinant. Later, we turn to ENPs as a robustness check.

The primary independent variable is district magnitude (dm), defined as the number of seats available in each district. For mayoral elections, district magnitude equals one, while for councillor elections it takes the values of six, eight, or ten. These values vary automatically according to population size, allowing us to observe institutional effects across a range of district magnitudes within the same electoral system.

Following the approach adopted by Singer and Gershman (2018), several contextual variables are included to account for district-level variation and broader national trends. One control is the total number of lists competing in each district, as ENPv is mechanically limited by the number of parties entering the race. Including this measure establishes a baseline for potential competition. This variable also serves as a proxy for potential electoral competition (Fiva and Folke 2016), capturing the range of parties that are able to enter the race under different institutional conditions. In addition, the national-level ENP is included to control for systemic changes or anomalies affecting all districts, such as the 2020 local election, which was postponed to 2021 due to the COVID-19 pandemic.

It is also important to account for the learning process of voters and parties, as suggested by Tavits (2005) and others (Bunker and Negretto

2024; Roussias 2022; Tavits and Annus 2006). This factor is particularly relevant because the first election in the series (2004) marked the debut of the separated-ballot concurrent local system. Prior to that, mayors and councillors were elected on the same ballot under a mixed-member arrangement. The effects of institutional variables may therefore have been less pronounced in the initial contests, as both voters and parties were adapting to new rules. To capture this adjustment, a dummy variable marks the first two elections in the series (2004 and 2008).

We also include a variable to account for the concurrent nature of local elections, capturing potential coattail effects between the two contests. This component has been included in comparable models since Cox (1997) and remains relevant for systems where simultaneous elections may transfer strategic incentives across tiers (Shugart and Carey 1992). In this case, the variable captures the possible influence of mayoral contests on councillor races, the main level of observation in this study.

Finally, a variable measuring ethnic fractionalisation at the district level is included to capture the influence of social diversity on party competition. According to theory, higher ethnic diversity should increase the effective number of vote-winning parties (ENPv), with its effect varying by district magnitude (Clark and Golder 2006; Teorell and Lindstedt 2010). This variable is entered both as a main effect and as an interaction term, following evidence that social heterogeneity can shape electoral competition even under majoritarian systems (Milazzo, Moser, and Scheiner 2018), and that the effect has been found to be present in the Chilean political system (Bunker 2025a).

5.2. *Methods*

To evaluate the causal impact of electoral reform on party system fragmentation, the study employs a difference-in-differences (DID) approach. This method compares variations in the effective number of vote-winning parties (ENPv) between districts that experienced an increase in the number of seats (treatment group) and those that did not (control group) over time. By controlling for pre-existing trends and other covariates, the DID framework isolates the effect of changes in district magnitude on party system fragmentation.

The model includes an interaction term between district magnitude and a treatment indicator that identifies districts experiencing a change in the number of seats. This specification captures the differential effect of institutional reform by comparing districts that increased their magnitude (from six to eight or from eight to ten seats) with those that remained constant. The dependent variable is ENPv, and the model also controls for election year and district-specific factors to account for

unobserved heterogeneity and time-related shocks. The model is specified as follows:

$$\text{ENPV}_{it} = \beta_0 + \beta_1 \times \text{dm}_{it} + \beta_2 \times \text{dm_change}_{it} + \beta_3 \times (\text{dm}_{it} \times \text{dm_change}_{it}) + \gamma_t + \delta_i + \varepsilon_{it}$$

where ENPV_{it} is the ENP in district i at time t , dm_{it} is the district magnitude, dm_change_{it} is the treatment indicator for changes in district magnitude, $\text{dm}_{it} \times \text{dm_change}_{it}$ is the interaction term capturing the differential effect of district magnitude changes, γ_t represents the election year fixed effects, δ_i represents the district fixed effects, and ε_{it} is the error term.

To further explore the impact of district magnitude on ENPv within a broader context, a log-linear mixed-effects model is also employed. This model extends the difference-in-differences approach by simultaneously accounting for variation across districts and over time. The specification includes both random and fixed effects. In the model, the logarithm of the ENP in district i at time t serves as the dependent variable, while the key predictor is the logarithm of dm in the same district and year. Random intercepts are included for both districts and election years to capture persistent differences and common temporal factors. The log-log structure assumes that the relationship between district magnitude and party fragmentation is proportional rather than linear, consistent with prior work showing that both ENP and district magnitude cannot take values of zero (Singer and Gershman 2018, 175). The coefficient on $\log(\text{dm})$ thus represents the percentage change in the ENP associated with a one-percent change in district magnitude. The model is specified as follows:

$$\log(\text{ENPV}_{it}) = \beta_0 + \beta_1 \times \log(\text{dm}_{it}) + u_i + v_t + \varepsilon_{it}$$

where $\log(\text{ENPV}_{it})$ is the ENP in district i at time t , $\log(\text{dm}_{it})$ is district magnitude, u_i and v_t are random effects for districts and years, and ε_{it} is the error term.

Figure 1 illustrates the distribution of districts by magnitude across the five local elections held between 2004 and 2021. The figure shows a steady increase in the number of districts with higher magnitudes, reflecting population growth over time. This gradual upward trend highlights the automatic adjustment mechanism in Chile's local electoral system, which incrementally increases district magnitude to preserve proportional representation as populations expand.

Table A1 in the Online Appendix presents the yearly evolution of district magnitudes between 2004 and 2021. It confirms that population growth, rather than political reform, was the main driver of institutional variation in district magnitude during this period.

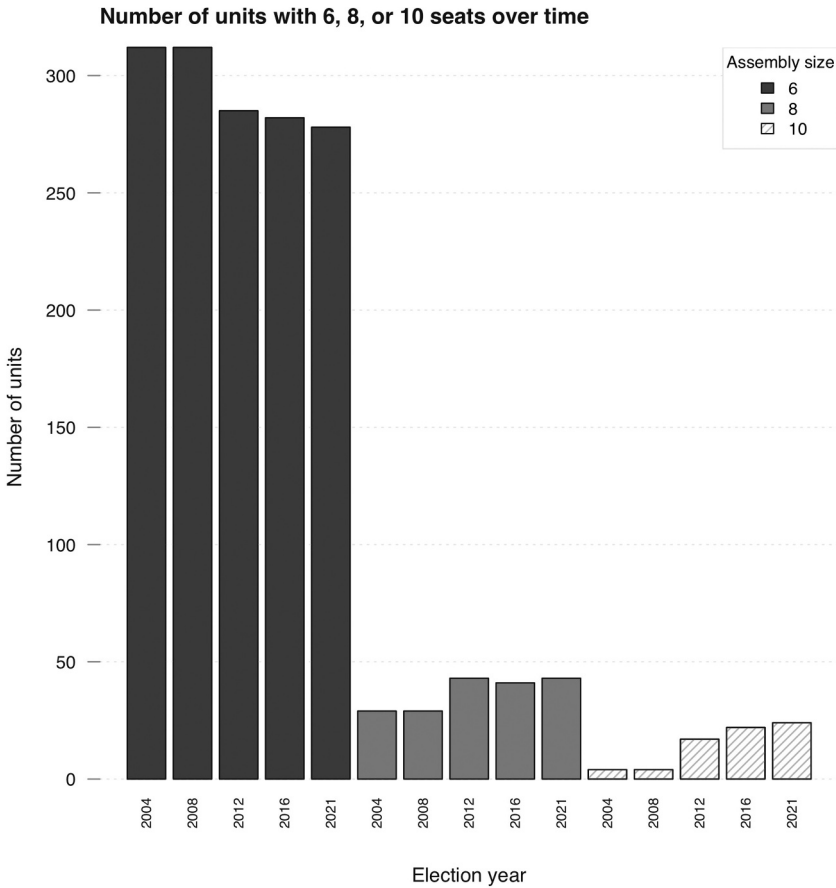


Figure 1. Evolution of district size in local councillor elections in Chile, 2004–2021.
Source: Authors.

6. Findings

Table 1 presents three DID specifications (M1–M3), which differ in their fixed-effects structure: M1 uses election fixed effects, M2 uses unit fixed effects, and M3 combines both.

In all three models (M1, M2, and M3), the coefficient for district magnitude is positive and statistically significant, indicating that increases in district magnitude are associated with higher values of ENPv. The coefficients — 0.36 in M1 ($p < 0.001$), 2.09 in M2 ($p < 0.001$), and 0.96 in M3 ($p < 0.001$) — confirm that larger districts, which offer more seats, foster greater party fragmentation by lowering the electoral threshold for smaller parties. These results directly support H1, which posits that increases in district magnitude lead to a higher ENPv.

Table 1. Difference-in-differences analysis of the impact of district magnitude on party fragmentation in Chilean local elections (2004–2021).

	[M1]	[M2]	[M3]
dm	0.36*** (0.05)	2.09*** (0.09)	0.96*** (0.09)
dm: dm 6 to 8	−0.03 (0.03)	−0.32*** (0.03)	−0.15*** (0.03)
dm: dm 8 to 6	0.12*** (0.01)	0.59*** (0.03)	0.36*** (0.03)
dm: dm 8 to 10	−0.09** (0.03)	−0.27*** (0.03)	−0.13*** (0.03)
2008	1.08*** (0.04)		1.08*** (0.04)
2012	1.07*** (0.05)		1.02*** (0.05)
2016	2.37*** (0.06)		2.22*** (0.07)
2021	3.32*** (0.09)		3.14*** (0.10)
345 units		Omitted	Omitted
Intercept	0.45 (0.30)	−8.00*** (0.53)	−2.76*** (0.533)
R2	0.599	0.368	0.775
Adj R2	0.597	0.208	0.718
Obs.	1,725	1,725	1,725

Dependent variable: Effective Number of Vote-Winning Parties (ENPv). Key: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are in parentheses.

The interaction terms clarify the effects of specific seat changes. The transition from six to eight seats shows negative coefficients in M2 (−0.32, $p < 0.001$) and M3 (−0.15, $p < 0.05$), suggesting that moderate seat increases yield limited additional fragmentation once thresholds are surpassed. The transition from eight to ten seats also produces negative coefficients in M1 (−0.09, $p < 0.01$) and M2 (−0.27, $p < 0.001$), indicating a plateauing effect at higher magnitudes. In the single case of a reduction from eight to six seats, the coefficient is positive and significant in M1 (0.12, $p < 0.001$) and M2 (0.59, $p < 0.001$), consistent with the expectation that smaller districts constrain the number of viable parties.

These results refine, rather than contradict, H1. While the overall coefficient for district magnitude is consistently positive, confirming that larger districts foster higher ENPv, the negative interaction terms for seat increases indicate that the marginal effect of additional seats diminishes once representation becomes highly permissive. In other words, when districts expand from six to eight or from eight to ten councillors, the entry barrier for new parties is already low, so the incremental gain in fragmentation becomes smaller. This pattern reflects a threshold or saturation dynamic typical of mature proportional systems, in which further expansions of magnitude yield proportionally fewer new competitors. The single positive coefficient

for the reduction from eight to six seats likely reflects local compositional or temporal effects in the lone district, rather than a reversal of the general institutional relationship.

The dummy variables for election years (2008, 2012, 2016, and 2021) control for time-specific effects. The coefficients are positive and statistically significant across models, indicating that the ENPv has generally increased over time, independent of changes in district magnitude. This pattern suggests a broader national trend towards greater party system fragmentation, reflecting long-term structural shifts in Chilean politics beyond the local institutional adjustments captured in the models.

Overall, the results in [Table 1](#) demonstrate that changes in district magnitude significantly affect the ENPv. Larger districts consistently lead to higher levels of party fragmentation, though the size of the effect varies with the magnitude of reform and contextual factors. Accounting for year effects (2008–2021) confirms that national fragmentation trends persist independently of local reforms. Collectively, these findings provide strong empirical support for H1 and validate the institutional prediction that permissive electoral arrangements increase the number of competitive parties.

[Figure 2](#), based on Model 3 from [Table 1](#), illustrates the average ENPv over time, distinguishing districts according to whether they experienced a change in district magnitude. The solid dark line represents districts with no change, the dashed line those that increased from six to eight seats, the dot-dash line those that increased from eight to ten, and the dotted line the single district that moved from eight to six. The figure shows that increases in district magnitude are associated with higher average levels of party fragmentation, particularly in districts that expanded from eight to ten seats, confirming that institutional adjustments promote greater competition among parties.¹

Districts that did not experience a change in district magnitude show a steady increase in ENPv over time, rising from around three in 2004 to slightly above five by 2021. This gradual upward trend suggests that even without institutional reform, there has been a general increase in party fragmentation. Districts that increased their magnitude from six to eight seats display a modest initial decline around 2008, followed by a pronounced rise reaching approximately eight by 2021. By contrast, districts that expanded from eight to ten seats show the sharpest growth in ENPv, particularly after 2016, approaching nine by 2021. Together, these trends confirm that larger district magnitudes are associated with greater party fragmentation, though natural increases over time also contribute to this evolution.

[Figure 3](#) provides a detailed examination of the evolution of the ENPv over time in districts that experienced particular changes in district magnitude. The figure is divided into two panels: the left panel displays districts that

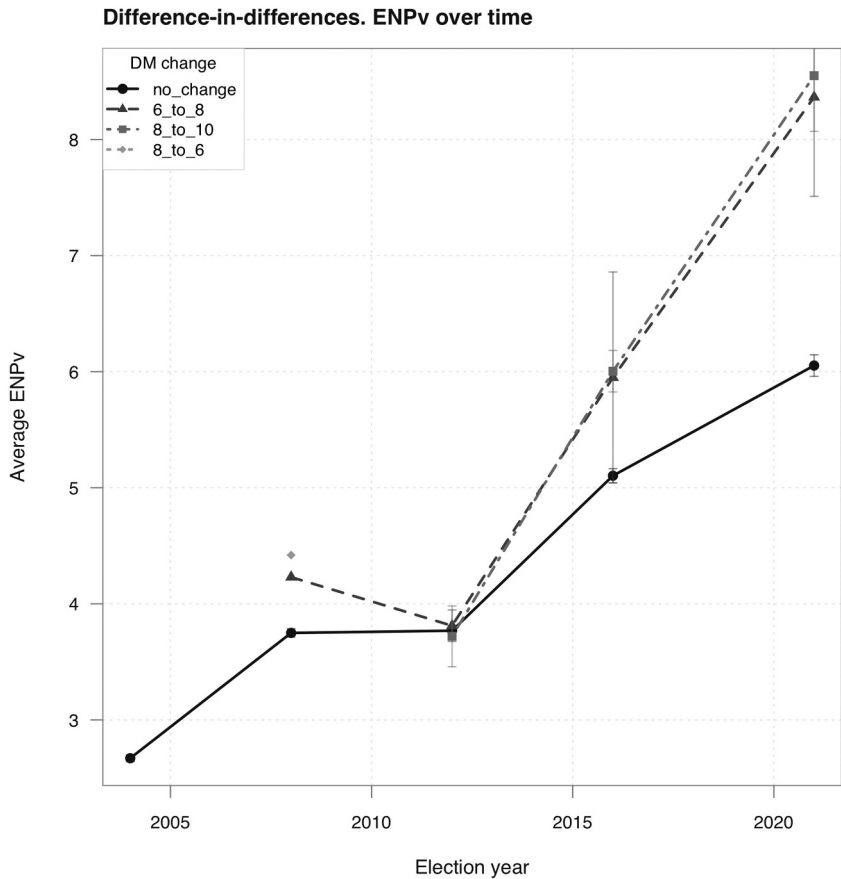


Figure 2. Average ENPv in groups of districts with changes to their district magnitude (Model 3, Table 1). Source: Authors.

increased from six to eight seats, while the right panel presents those that expanded from eight to ten seats. This design facilitates direct comparison of the magnitude and timing of institutional effects across different types of reform.

Comparing the two panels in Figure 3 shows that districts increasing from eight to ten seats experience a more pronounced and consistent rise in ENPv than those moving from six to eight. This pattern confirms that larger institutional changes – at higher scales of district magnitude – tend to have stronger effects on party system fragmentation. In both panels, the steepest increases in ENPv occur after 2016, suggesting that time-specific political or social factors during this period may have amplified the effects of institutional reform. At the same time, the figure reveals a flattening of the slope at higher magnitudes, consistent with the diminishing marginal effects identified in

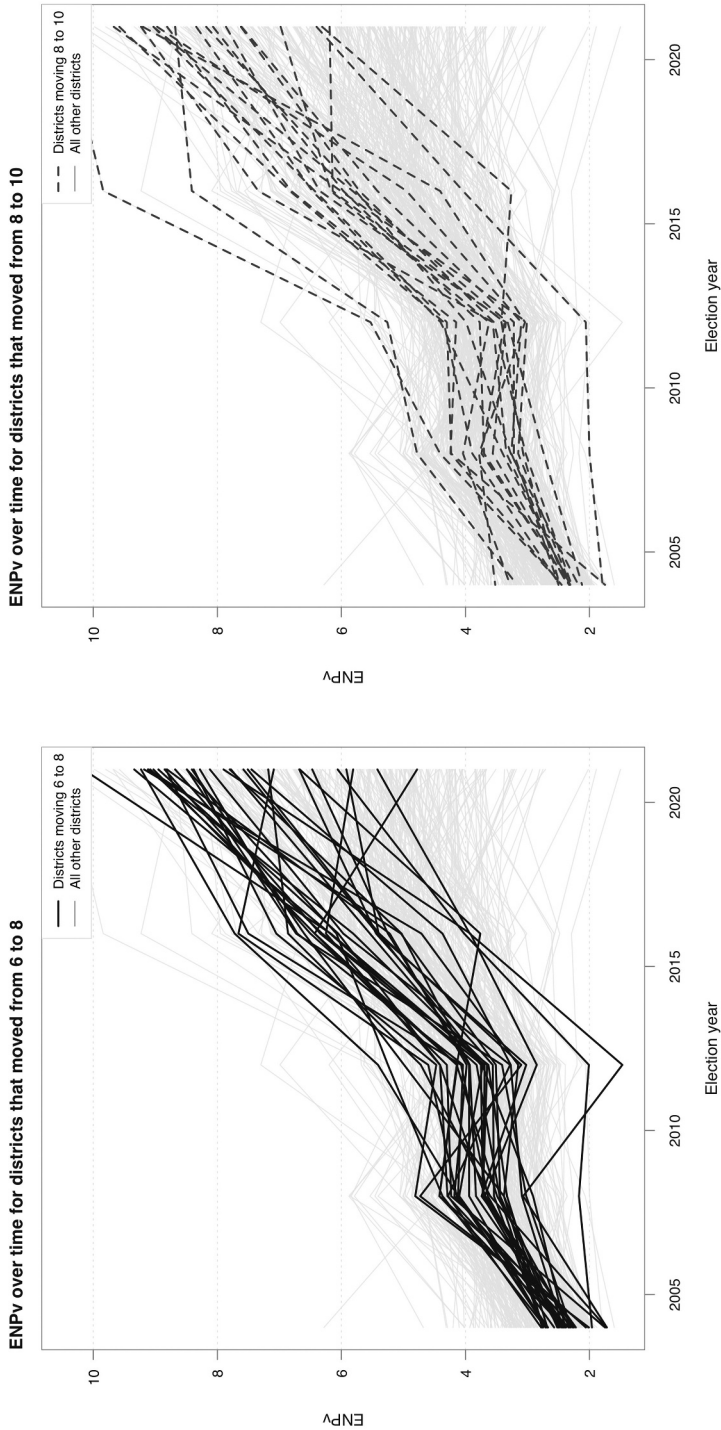


Figure 3. Evolution of effective number of vote-winning parties in Chilean districts experiencing changes in district magnitude. Source: Authors.

Table 1: while increases in district size continue to raise fragmentation overall, the rate of increase slows as representation becomes more permissive. This plateau effect visually reinforces the threshold or saturation dynamic noted earlier.

Across all districts, the overall upward trend in ENPv is evident, indicating that party fragmentation generally rises as district magnitude expands. However, the trajectories of individual districts, represented by the individual lines, display considerable variation. Some districts show sharp increases, while others change more gradually, implying that the effects of institutional reform are mediated by district-specific conditions such as demographic composition, political competition, or historical voting patterns.

These findings underscore the complex relationship between district magnitude and party fragmentation and highlight the need for analytical approaches capable of capturing this heterogeneity. The observed variation across districts and over time points to the influence of both contextual and temporal dynamics. To account for these factors, the following section employs multilevel models that incorporate fixed and random effects, accommodating the hierarchical structure of the data and isolating both district-level and time-specific influences.

Table 2 presents the results from a series of mixed-effects models incorporating both random and fixed effects, with the ENPv as the dependent

Table 2. Determinants of party system fragmentation across districts in Chilean local elections: mixed-effects model analysis.

	[M4]	[M5]	[M6]	[M7]	[M8]	[M9]
log(dm)	0.50*** (0.05)	0.37*** (0.05)	0.37*** (0.05)	0.37*** (0.05)	0.37*** (0.05)	0.34** (0.10)
log(ENPv) for Mayoral Elections		0.31*** (0.02)	0.31*** (0.02)	0.31*** (0.02)	0.31*** (0.02)	0.31*** (0.02)
First Two Elections			−0.40* (0.20)	−0.02 (0.10)	−0.02 (0.10)	−0.02 (0.10)
log(ENPv National)				0.51*** (0.11)	0.51*** (0.11)	0.51*** (0.11)
Ethnic Fractionalisation					−0.03 (0.04)	−0.37 (0.88)
log(dm) * Ethnic Fractionalisation						0.19 (0.49)
Intercept	0.47** (0.17)	0.42** (0.16)	0.58*** (0.15)	−0.37 (0.22)	−0.36 (0.22)	−0.29 (0.28)
Random Effects						
Sigma2	0.03	0.03	0.03	0.03	0.03	0.03
t00 [unit]	0.02	0.01	0.01	0.01	0.01	0.01
t00 [election]	0.09	0.08	0.05	0.01	0.01	0.01
ICC	0.76	0.75	0.65	0.33	0.33	0.33
N [unit]	345	345	345	345	345	345
N [election]	5	5	5	5	5	5
Obs.	1,725	1,725	1,725	1,725	1,725	1,725
Marginal R2/ Conditional R2	0.029 / 0.769	0.081 / 0.773	0.388 / 0.784	0.657 / 0.770	0.658 / 0.770	0.658 / 0.769

Dependent variable: Effective Number of Vote-Winning Parties (ENPv). Key: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are in parentheses.

variable. The table reports six specifications (M4 through M9), in which variables are added sequentially to progressively expand the analysis and assess the robustness of results.

M4 begins with a basic specification including only the logarithm of district magnitude ($\log dm$). M5 adds the $\log ENP_v$ for mayoral contests to capture concurrency effects. M6 adds a dummy variable for the first two elections in the series to account for potential anomalies related to early system adjustment. M7 includes the national-level effective number of parties ($\log$) to control for broader political trends. M8 and M9 add a control for ethnic fractionalisation and its interaction with $\log(dm)$, both proving statistically insignificant ($p > 0.1$), confirming that institutional variables explain the bulk of variation in fragmentation.

M4 indicates that the institutional theory holds strongly at the district level in Chile's local elections. The coefficient for $\log(dm)$ is 0.50 ($p < 0.001$), meaning that a one-percent increase in district magnitude is associated with a 0.50-percent rise in ENP_v . In practical terms, an increase from six to eight or from eight to ten seats is associated with roughly one additional effective vote-winning party, on average. In all extended models (M5–M9), the effect of district magnitude remains positive and significant, while the marginal R^2 increases from 0.03 to 0.66 as contextual factors are added, indicating that added explanatory power operates at the fixed-effects level. The coefficient for the ENP in mayoral contests ($\log ENP_v \text{ mayor} = 0.31$, $p < 0.001$) confirms H2: concurrent FPTP elections for mayors amplify vote-level fragmentation. Variance-inflation factors below 2.5 indicate no multicollinearity concerns.

The findings highlight the central role of district magnitude and concurrency in shaping party fragmentation at the local level. While the variable for ethnic fractionalisation remains statistically insignificant as expected, the consistently positive relationships between district magnitude, mayoral concurrency, and ENP_v strongly support the institutional explanation of fragmentation. National-level effects further underscore the integration of local and national party systems.

To verify the robustness of these findings, the analysis next examines the determinants of the ENPs. [Table 3](#) retains the variables used in the previous models but adds the total number of lists competing in each district and ENP_v as predictors to provide a baseline and to account for the mechanical relationship between vote and seat fragmentation (Blais and Carty 1991). The variables related to ethnic fractionalisation are excluded, as social structures are theorised at the national rather than district level.

The results provide additional support for district magnitude as a robust determinant of party-system fragmentation. Across all models (M10–M14), the coefficient for $\log(dm)$ remains positive and significant, ranging from 0.39 to 0.17, all significant at $p < 0.001$. This means that a 33% increase in

Table 3. Robustness analysis: determinants of the effective number of seat-winning parties (ENPs) across districts.

	[M10]	[M11]	[M12]	[M13]	[M14]
log(d)	0.39** (0.05)	0.17*** (0.05)	0.23*** (0.03)	0.23*** (0.03)	0.23*** (0.03)
Number of Lists		0.04*** (0.00)	-0.07*** (0.00)	-0.07*** (0.00)	-0.07*** (0.00)
log(ENPv)			1.16*** (0.02)	1.16*** (0.02)	1.16*** (0.02)
log(ENPv) for Mayoral Elections				0.01 (0.01)	0.01 (0.01)
First Two Elections					-0.00 (0.01)
Ethnic Fractionalisation					
log(dm) * Ethnic Fractionalisation					
Intercept	0.39*** (0.15)	0.53*** (0.12)	-0.49*** (0.05)	-0.50*** (0.05)	-0.49*** (0.05)
Random Effects					
Sigma2	0.04	0.04	0.02	0.02	0.02
t00 [unit]	0.01	0.01	0.00	0.00	0.00
t00 [election]	0.06	0.04	0.00	0.00	0.00
ICC	0.62	0.51	0.01	0.01	0.01
N [unit]	345	345	345	345	345
N [election]	5	5	5	5	5
Obs.	1,725	1,725	1,725	1,725	1,725
Marginal R2/ Conditional R2	0.022 / 0.626	0.102 / 0.556	0.852 / 0.853	0.852 / 0.853	0.852 / 0.853

Dependent variable: Effective Number of Seat-Winning Parties (ENPs). Key: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are in parentheses.

magnitude – roughly two additional seats – raises ENPs by about 13%, a more modest response than that observed at the vote level. These findings reinforce H1: district magnitude continues to drive party fragmentation, although its influence on seats is smaller than on votes, consistent with mechanical conversion effects (Fiva and Folke 2016; Singer and Gershman 2018).

This mechanical relationship, between vote-winning and seat-winning parties, is also evident in Models 12 through 14, after ENPv is entered as a predictor to isolate the effect of district magnitude on ENPs. The association remains positive and statistically significant, even when the number of registered lists is included as a control. As the proxy for potential competition fluctuates, the coefficient for ENPv remains consistently around 1.16 across these models, reinforcing the importance of controlling for the number of relevant parties rather than simply counting the total number of competing lists.

Taken together, the difference-in-differences and mixed-effects analyses confirm both hypotheses. Increases in district magnitude raise vote- and seat-level fragmentation (H1), while concurrent FPTP mayoral elections heighten vote-level competition without producing equivalent seat-level fragmentation (H2). These results demonstrate that institutional design – rather than

social or contextual factors – accounts for most observed variation in party-system fragmentation across Chilean districts.

Model diagnostics, including residual and collinearity checks, suggest stable and well-specified models. Random-effects estimates remain consistent across specifications, indicating that the reported effects are not driven by multicollinearity or omitted-variable bias.

7. Discussion

The analysis strongly supports both hypotheses. For the first hypothesis, concerning the effect of district magnitude on ENPv, the data consistently show that increases in district magnitude are associated with a significant rise in the ENPv. Across models, the coefficient for district magnitude ($\log(dm)$) remains positive and significant. For instance, in Model 10, a 33% increase in district magnitude – such as an increase from six to eight seats – is expected to result in approximately a 13% increase in ENPv. This finding aligns with institutional theory, which posits that larger districts, offering more seats, lower the electoral threshold for smaller parties and thereby foster greater party fragmentation (Shugart and Taagepera 2017; Taagepera 2007). The consistent significance of this relationship across multiple models reinforces the robustness of the institutional approach (Li and Shugart 2016).

Importantly, the findings also indicate that the effect of district magnitude is not linear. While increases in magnitude consistently raise ENPv, the marginal impact of additional seats diminishes at higher levels of permissiveness. Once district magnitude surpasses key representational thresholds, further increases lower entry barriers less dramatically, producing a saturation dynamic rather than proportional growth in fragmentation (see Carey and Hix 2011).

The temporal dimension of institutional effects also warrants attention. The dummy variable identifying the first two elections in the series (2004 and 2008) captures the adjustment period during which voters and parties were still learning to coordinate under the new concurrent local electoral framework. Its inclusion in the models controls for this initial period of lower institutional alignment, consistent with Taagepera's (2002) expectation that electoral outcomes converge towards institutional predictions only after several cycles. The results suggest that fragmentation levels in the first two elections were indeed lower than in subsequent contests, supporting the view that the full effect of institutional incentives materialises gradually rather than immediately – a pattern previously observed (Bunker and Negretto 2024; Maškarić 2026; Roussias 2022; Tavits and Annus 2006).

For the second hypothesis, concerning the conditioning effect of concurrent first-past-the-post (FPTP) elections, the data provide further supporting evidence. Models that control for simultaneous FPTP mayoral contests

confirm that the presence of FPTP elections conditions behaviour in proportional representation (PR) council elections. Concurrent FPTP elections affect how voters and parties behave in PR council elections, increasing vote fragmentation but limiting its translation into seat fragmentation (Calvo and Hellwig 2011; Riera 2013). When the ENPv for mayoral elections is included as a predictor, the coefficient remains consistently significant and stable, indicating that concurrency shapes vote-level competition in the PR tier (Cox 1990).

At the same time, the analysis reveals clear limits in how vote-level fragmentation translates into seats. Although concurrent FPTP elections increase ENPv, this heightened competition does not convert proportionally into higher ENPs at the seat-allocation stage. This reflects conversion limits under proportional representation: mechanical allocation rules and effective thresholds absorb a portion of vote fragmentation, preventing a one-to-one translation into legislative representation (Fiva and Folke 2016; Singer and Gershman 2018). In this sense, PR moderates but does not eliminate the centrifugal effects generated at the vote level.

By contrast, there is no evidence that social structure – operationalised through ethnic fractionalisation – directly affects party fragmentation at the district level. This differs from findings in established democracies (Harfst et al. 2018; Potter 2014), but it may simply reflect that such indices capture little variation at subnational levels or are redundant in the Chilean context. This outcome is consistent with recent accounts suggesting that the institutional and contextual dimensions of party competition outweigh the effects of social heterogeneity in shaping party systems (Moser, Scheiner, and Stoll 2018; Stoll 2008), although when measured at the district-level, its effect may be dependent on additional variables, such as the number of candidates in competition (Bunker 2025a) or territorial concentration (Bochsler 2011).

In summary, the evidence provides strong empirical support for both hypotheses. The institutional impact of district magnitude on party fragmentation is clear, with larger districts generating a higher ENP. The conditioning effect of concurrent FPTP elections is also confirmed, demonstrating that the design and interaction of electoral rules significantly shape party system outcomes.

These findings carry implications that extend beyond the Chilean case. For new democracies more broadly, the results demonstrate that institutional design can exert an independent and identifiable effect on party-system fragmentation even in contexts of frequent reform and incomplete democratic consolidation (Mainwaring and Zoco 2007; Moser and Scheiner 2012). The diminishing marginal returns documented here refine the institutional prediction: permissive rules consistently raise fragmentation, but the relationship is not linear, and further expansion of magnitude yields proportionally smaller gains once representation becomes highly permissive. The

conversion limits identified under concurrent FPTP and PR rules similarly suggest that vote-level and seat-level fragmentation must be analysed separately, as mechanical allocation rules absorb a portion of competitive incentives generated at the ballot stage. Together, these patterns suggest that institutional theories developed in consolidated democracies travel to new democracy contexts (Lucardi 2019; Maškarinec 2026), but require qualification – particularly regarding saturation dynamics (consistent with Lago and Martínez Coma 2024) and the temporal pace at which institutional effects materialise.

Beyond institutional theory, these findings speak to local government as a substantive object of inquiry. Kouba and Dosek (2022) document a Latin American tradeoff in which smaller municipalities raise participation but lower contestation. The results here show that the same population thresholds that automatically expand council magnitude also enlarge the field of effective parties: contestation is not exogenous to municipal size but institutionally amplified by it. Two implications follow. First, further expansions in council size beyond ten seats are likely to yield smaller gains in the representational breadth of local councils. Second, concurrency with the FPTP mayoral race structures how voters bundle accountability across the executive and legislative tiers of the commune – a consideration for the wider machinery of Chilean local government, where transparency, participation and accountability remain active sites of research (Disi Pavlic and Rebolledo Toro 2022; Hernández-Bonivento and Moller 2025).

8. Concluding remarks

The study of electoral reform in new democracies is essential because these political systems are often fluid and experience frequent institutional change. Understanding how such reforms influence the ENP provides critical insight into broader processes of democratic consolidation and stability. Chile offers a valuable case for analysing these dynamics, as recurrent adjustments in district magnitude and concurrent electoral rules have directly shaped patterns of party competition and representation.

The combination of a difference-in-differences framework and mixed-effects models demonstrates the analytical value of district-level data for isolating causal relationships in complex institutional environments. Leveraging subnational data not only mitigates the contamination of national trends and temporal shocks but also captures the local mechanisms through which institutional change alters competition. Together, these contributions speak both to institutional theory in new democracies and to local government as a substantive object of inquiry in its own right.

Future research could extend this framework in two directions. Comparatively, applying it across other new democracies would test whether

the patterns identified here – diminishing marginal returns at higher magnitudes, vote-to-seat conversion limits, and the gradual materialisation of institutional effects – generalise beyond Chile. Within Chile, future work could examine whether similar effects emerge in other electoral arenas once population-based seat adjustments become available, such as in elections for deputies or regional councillors, and whether district magnitude interacts with factors such as party-system nationalisation or fiscal decentralisation. Longitudinal analyses following future census updates would also allow assessment of whether the effects of institutional reform persist, diminish, or compound over time.

Note

1. The dot represents the lone district that decreased in magnitude from 8 to 6 seats. The error bars represent the confidence intervals for the average ENPv in each group. The relatively small size of these intervals in most cases indicates a high level of precision in the estimates. However, the slight widening of error bars in the later years, especially in 2021, suggests increased variability in party fragmentation outcomes as the district magnitude changes take effect.

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